Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

pmsimstats team

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1 Abstract

Background. Aggregated N-of-1 clinical trials test biomarker-treatment interactions through a fixed continuous- heterogeneity regression in published practice. The psychometric literature has developed mature finite-mixture and latent-class machinery for representing discrete subpopulation structure that the biostatistical literature on N-of-1 trials has not engaged. The choice between continuous-heterogeneity and latent-class encodings of the biomarker-moderation question is a substantive scientific question rather than a parameterisation convenience, but the field has not characterised when class-aware analyses become competitive with continuous regression at trial-relevant sample sizes.

Methods. We develop a five-component biomarker-moderation spectrum (covariance-only, mean-and-covariance, location-scale, heterogeneous random-effects, latent-class) bridging continuous-heterogeneity and latent-class formulations, and we catalogue the finite-mixture taxonomy (latent class analysis, growth mixture models, factor mixture models, regression mixtures) for application to within-subject longitudinal designs. We commit a heterogeneous-random-slopes generative DGP and a class-aware analysis pipeline (`lcmm::hlme` with gridsearch initialisation), and run a 240-replicate-per-cell prototype simulation at $N = 70$, gating slope $\in \{0, 1.0, 1.5\}$, on a 4-core parallel implementation. Three test forms are recorded per replicate: the linear-mixed `bm:Dbc` Wald, the unconditional Wald on the lcmm gating coefficient, and a BIC-conditional gating procedure.

Results. In a 240-replicate-per-cell pilot we find that the unconditional Wald

test on the latent-class gating coefficient inflates Type I error to 0.197 (95% CI $[0.145, 0.250]$) at the null, decisively above nominal 0.05 and consistent with the[1] and[2] overextraction predictions. The BIC-conditional gating procedure controls Type I (0.004 under the null) but is severely under-powered at trial-relevant N (rejection rate 0.025 at gating slope 1.0, 0.004 at 1.5). The linear-mixed `bm:Dbc` Wald test holds nominal Type I and reaches 0.542 power at gating slope 1.5, dominating both class-aware tests. MCSE on power is approximately 0.030 across the grid. These pilot estimates precede the production five-study programme described in the companion simulation plan, which has not yet been executed.

Conclusions. Within the 240-replicate pilot, at trial- relevant sample sizes ($N = 70$, eight measurement timepoints), neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline under the heterogeneous-random-slopes DGP. The latent-class encoding appears to require a regime of large class separation for class-aware tests to gain power; the pre-registered production programme (five studies, $N \in \{35, 70, 100\}$, wider gating- slope grid) has not yet been executed and is required to map the regime where the BIC-conditional procedure becomes competitive.

2 Introduction

2.1 The aggregated N-of-1 trial and the biomarker-moderation question

The aggregated N-of-1 design pools within-person treatment contrasts across participants for population-level inference.

- Each participant cycles between active treatment and control across multiple within-person blocks.
- The design outperforms parallel-group trials in information per participant when treatment response is reversible on a useful timescale.
- Deployed in chronic pain, neuropsychiatric PTSD, and rare-disease settings where parallel-group trials are infeasible.

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The aggregated N-of-1 trial design is a class of clinical trials in which each enrolled participant cycles between active treatment and control across multiple within-person blocks, with within-person comparisons pooled across participants for population-level inference [@guyatt1986; @guyatt1988; @lillie2011; @senn2018]. The design preserves the within-subject contrast that gives single-participant N-of-1 trials their inferential strength while accumulating the statistical efficiency of a multi-participant trial. For chronic conditions whose *treatment response is reversible on a useful timescale, the design yields more information per participant per unit time than a parallel-group randomised trial of equivalent total sample size [@mirza2017]. It has been deployed in chronic pain [@yelland2009], inflammatory bowel disease, neuropsychiatric symptoms in post-traumatic stress disorder [@hendrickson2020], and a growing portfolio of rare and orphan disease applications where parallel-group trials are infeasible [@lillie2011; @schork2015].*

The primary scientific estimand in biomarker-guided N-of-1 trials is the biomarker-by-treatment interaction term.

- A predictive biomarker enables point-of-care treatment stratification per the PATH framework.
- Power depends on moderation magnitude, within-participant contrast precision, trial design, and carryover leakage.
- Carryover erodes the cleanness of within-subject contrasts and is the principal nuisance parameter in power calculations.

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Within this design class, a question of increasing scientific weight is whether treatment response is moderated by a measured pre-treatment biomarker. A biomarker that predicts which participants respond to which treatments would allow clinicians to stratify treatment decisions at the point of care, and the formal framework for such inference is laid out in the PATH statement [@kent2020path]. The corresponding statistical estimand is the biomarker-by-treatment interaction term in the within-participant outcome model. Power to detect this interaction depends on the magnitude of moderation, the precision of the within-participant contrasts, the trial design (open-label, crossover, hybrid), and any time-dependent leakage between treatment phases, conventionally termed 'carryover', that erodes the cleanness of the within-subject contrasts.

The DGP encoding of biomarker moderation determines carryover-induced power loss, a question the prior literature raised but did not resolve.

- Power loss under carryover varies sharply with how moderation is encoded in the DGP.
- Two formal answers from outside biostatistics – continuous heterogeneity and latent-class structure – dominate the available options.
- This paper maps those traditions and their consequences for N-of-1 trial design and analysis.

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Recent methodological work has documented substantial power losses for biomarker-by-treatment interaction inference under realistic carryover, with the loss varying sharply by how the moderation effect is encoded in the data-generating process [@hendrickson2020]. The present paper takes up a question that the prior work raised but did not resolve: how should biomarker moderation be represented in the DGP itself? Two formal answers, drawn from a long history of work outside biostatistics, dominate the available options. We begin by laying out these two answers and the methodological tradition behind each, before turning to the specific consequences for N-of-1 trial design and analysis.

2.2 Two formal answers: continuous heterogeneity and latent-class structure

The continuous-heterogeneity answer encodes biomarker moderation as a smooth interaction in a single multivariate distribution.

- Every participant has some treatment response; the response is a linear function of the biomarker.
- Population heterogeneity is captured by a single distribution with an interaction parameter $\beta_{bm:D}$.
- This is the dominant tradition in clinical biostatistics via random slopes and fixed-effects interaction terms.

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The first answer represents biomarker moderation as a graded continuous effect: every participant has some degree of treatment response, the response is a smooth (typically linear) function of the biomarker, and population heterogeneity is captured by a single multivariate distribution with an interaction parameter. This is the dominant tradition in clinical biostatistics, where heterogeneity of treatment effect is routinely modelled via random slopes on the treatment indicator and via interaction terms in fixed-effects regression. In its most economical form, the moderation enters the conditional mean of the outcome as $\beta_{bm:D} \cdot B \cdot D$, where B is the biomarker and D is the (possibly continuous-decay) treatment indicator.

The latent-class answer encodes biomarker moderation as discrete subpopulation structure, with the biomarker predicting class membership.

- Participants belong to unobserved responder/non-responder classes with qualitatively distinct treatment-response curves.

- Class membership is a Bernoulli draw with biomarker-dependent mixing probabilities.
- Marginal moderation emerges from the mixture, not a smooth conditional-mean function.

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The second answer represents biomarker moderation as a discrete subpopulation structure: participants belong to one of a small number of unobserved classes (most commonly 'responder' and 'non-responder'), the classes have qualitatively distinct treatment-response curves, and the biomarker predicts class membership without determining it. The corresponding generative form is a finite mixture in which class membership is a Bernoulli or multinomial draw with mixing probabilities that depend on the biomarker, and the within-class outcome model is class-specific. The marginal moderation observed in the data emerges from the mixture rather than from a smooth conditional-mean function.

The two encodings reflect genuinely distinct biological hypotheses that diverge under the longitudinal structure of an N-of-1 trial.

- Continuous heterogeneity implies a graded physiological substrate scaling drug effect; latent-class implies qualitatively distinct phenotypes.
- Both views yield identical marginal associations at a single timepoint but diverge once carryover and serial dependence are engaged.
- Methodology from outside biostatistics – with mature formalisation and identifiability analysis – is directly applicable.

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These two answers correspond to genuinely different biological hypotheses. The continuous-heterogeneity view is consistent with a mechanism in which the biomarker reflects a graded physiological substrate (receptor density, enzymatic activity, baseline trait severity) that scales drug effect across the entire population. The latent-class view is consistent with a mechanism in which qualitatively distinct phenotypes exist (a metabolising versus non-metabolising allelic variant; a present versus absent target receptor) and the biomarker is an imperfect indicator of phenotype. The two views can produce identical marginal biomarker-outcome associations at a single timepoint and diverge sharply once the longitudinal structure of an N-of-1 trial is engaged, because *carryover and within-person serial dependence interact differently with continuous and class-structured heterogeneity. This is the substantive point at which methodology imported from outside biostatistics becomes useful: both traditions have long histories of formalisation and identifiability analysis in their respective fields, and the corresponding tools are mature.*

2.3 Psychometric origins

The continuous-trait tradition traces to Spearman and the factor-analytic lineage of Thurstone and Lawley, in which heterogeneity is by assumption continuous.

- Observed indicators are modelled as linear combinations of unobserved continuous factors plus residual noise.
- Between-person heterogeneity is summarised by factor scores along one or more dimensions.
- This tradition forms the psychometric foundation for the continuous-heterogeneity encoding of biomarker moderation.

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The continuous-trait view in formal statistical work originates with Spearman's two-factor theory of intelligence [@spearman1904], in which performance on cognitive tests was modelled as the sum of a single general factor and test-specific factors. Spearman's machinery was elaborated through the 1930s and 1940s by Thurstone, Hotelling, Lawley, and others into the modern factor-analytic tradition, in which observed indicator variables are written as linear combinations of unobserved continuous factors plus residual noise [@thurstone1947; @lawleymaxwell1971]. Within this tradition, between-person heterogeneity in any latent construct is by assumption continuous, and individual differences are summarised by factor scores along one or more dimensions.

The latent-class tradition originates with Lazarsfeld's local- independence formulation and was placed on rigorous ML footing by Goodman, then extended to longitudinal and covariate-dependent forms.

- Local independence within classes is the founding assumption of LCA; Goodman derived identifiability conditions and ML estimation.
- Extensions through the 1970s–1980s cover polytomous and continuous indicators, multilevel data, and covariate-dependent class probabilities.
- McLachlan and Peel's monograph is the standard entry point for finite mixture models.

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The discrete-class view emerged from a parallel but methodologically distinct literature beginning with Lazarsfeld's work on latent structure analysis in the 1950s [@lazarsfeld1950; @lazarsfeldhenry1968]. Lazarsfeld observed that in attitudinal and survey data, the joint distribution of categorical responses to multiple items frequently displayed structure inconsistent with a single continuous latent trait, and instead suggested that respondents belonged to one of several qualitatively distinct latent classes within which item responses were locally independent. This 'local independence' formulation is the founding assumption of latent class analysis. Goodman [-@goodman1974] placed the LCA model on rigorous statistical footing, deriving identifiability conditions and maximum-likelihood estimation procedures for the binary-indicator case in a Biometrika paper that remains a standard reference. Through the 1970s and 1980s, the LCA framework expanded to polytomous and continuous indicators, the latter widely known as latent profile analysis [@mccutcheon1987], to multilevel and longitudinal extensions [@vermunt2003], and to mixtures with covariate-dependent class probabilities [@daytonmacready1988; @bandeenroche1997]. McLachlan and Peel [-@mclachlanpeel2000] unified much of this material into a comprehensive monograph on finite mixture models that is the standard entry point for quantitative researchers from outside the discipline.

The psychometric literature has never resolved the continuous-versus-discrete question, motivating hybrid factor mixture models.

- Bartholomew argued the data rarely settle the continuous-discrete choice without strong prior assumptions.
- Cognitive ability favours continuous factors; certain psychopathologies favour discrete subtypes.
- Yung and Arminger developed factor mixture models combining discrete classes with within-class continuous factors.

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The defining tension in psychometric methodology has been whether a given pattern of observed heterogeneity is best understood as continuous or as discrete. Bartholomew [-@bartholomew1987] framed this as the choice between 'graded' and 'unfolding' models of individual differences and argued that the data themselves rarely settle the question without strong prior assumptions. The empirical literature has accumulated examples in both directions: cognitive ability is generally well-described by continuous factor models, while *certain forms of psychopathology display latent-class structure consistent with qualitatively distinct subtypes. Subsequent methodological work, beginning with @yung1997 and @arminger1999, developed hybrid frameworks (factor mixture models) in which discrete classes coexist with within-class continuous factors. We return to these frameworks below.*

2.4 Econometric and marketing-science development

Heckman and Singer's econometric work on unobserved heterogeneity anticipated finite-mixture methodology, warning that structural estimates are sensitive to the assumed heterogeneity distribution.

- The econometric concern was unobserved heterogeneity in micro-econometric models where identical-agent assumptions fail.
- Heckman and Singer proposed non-parametric ML with the heterogeneity distribution estimated as a discrete mixture.
- Lindsay unified the theory of mixture likelihoods; this work directly informed biostatistical finite-mixture practice.

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A largely independent strand of work on finite mixtures developed in econometrics and quantitative marketing through the 1980s and 1990s. The motivating problems differed from psychometrics: rather than measurement of cognitive or psychological constructs, the econometric concern was unobserved heterogeneity in micro-econometric models (consumer choice, labour-market participation, firm productivity) where standard regression assumptions of identical agents were known to fail. Heckman and Singer [-@heckmansinger1984], in an Econometrica paper on duration models, demonstrated that estimates of structural parameters in survival-style models could be highly sensitive to the assumed functional form of the unobserved heterogeneity distribution, and recommended a non-parametric maximum-likelihood approach in which the heterogeneity distribution itself was estimated as a discrete mixture with unknown support points and weights. Their estimator anticipated much of what would later become standard finite-mixture methodology in biostatistics, and Lindsay [-@lindsay1995] placed the broader theory of mixture likelihoods on a unified mathematical footing.

The marketing-science literature formalised covariate-dependent gating functions and clusterwise regression, providing the vocabulary now used in mixture-of-experts models.

- DeSarbo and Cron introduced clusterwise linear regression with cluster-specific coefficients; Wedel and DeSarbo extended it to GLMs.
- The gating function – class-membership probability as a function of covariates – became a first-class object.

- Jacobs et al. formalised the same structure as 'mixtures of experts' in machine learning, supplying the standard vocabulary.

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In quantitative marketing, the parallel concern was customer segmentation. A population of consumers might respond to a marketing intervention differently depending on which unobserved segment they belonged to, and the firm needed both to estimate segment-specific response functions and to predict segment membership for new consumers. DeSarbo and Cron [-@desarbocron1988] formalised this as 'clusterwise linear regression', in which observations were partitioned into clusters with cluster-specific regression coefficients. Wedel and DeSarbo [-@wedeldesarbo1995] generalised the approach to generalised linear models, and Jedidi and colleagues [-@jedidi1997] extended it to structural equation models with unobserved segments. Within this literature, the gating function (the model for the probability of segment membership as a function of covariates) became a first-class object, and methodological work focused on whether the gating model could share covariates with the within-segment regression model. Jacobs and colleagues [-@jacobs1991], working contemporaneously in machine learning, formalised an essentially identical structure under the name 'mixtures of experts', which provided much of the now-standard vocabulary.

Two econometric themes translate directly to the N-of-1 trial setting: identifiability concerns at small N, and biomarker-as-gating- covariate structure.

- Econometric identifiability concerns are acute when heterogeneity is the principal inference target rather than a nuisance.
- The biomarker-treatment moderation question is whether the gating function depends on the biomarker and within-class slopes differ.
- These two themes directly motivate the present paper's framing.

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Two themes from the econometric strand are particularly relevant to the N-of-1 trial setting. First, the econometric tradition treated unobserved heterogeneity as a nuisance in many applications, and was correspondingly attentive to whether the heterogeneity distribution was identified, weakly identified, or over-fit. The identifiability concerns raised by @heckmansinger1984 *and elaborated by @lindsay1995 translate directly into N-of-1 settings where sample sizes are small and the heterogeneity is the principal object of inference rather than a nuisance. Second, the marketing-science focus on covariate-dependent gating functions corresponds exactly to the biomarker-as-class-membership-predictor structure that motivates the present paper. The biomarker-treatment moderation question, translated into econometric vocabulary, is whether the gating function depends on a single observed covariate (the biomarker) and whether the within- class regression coefficient on treatment differs across classes.*

2.5 Convergence on hybrid frameworks

Psychometric and econometric traditions converged on factor mixture models combining discrete class structure with within-class continuous heterogeneity.

- Yung developed the FMM as a finite mixture of confirmatory factor models with class-specific loadings and residual covariances.
- Arminger placed FMMs in the Mplus structural equation framework.
- Lubke and Muthen provided the first systematic Monte Carlo evaluation of conditions under which class structure is distinguishable from continuous heterogeneity.

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By the late 1990s, both the psychometric and the econometric strands had converged on a hybrid formulation: combine continuous within-class factor structure with discrete between-class structure. Yung [-@yung1997], in Psychometrika, developed the factor mixture model as a finite mixture of confirmatory factor models, where each class had its own factor-loading and residual covariance structure. Arminger and colleagues [-@arminger1999] placed factor mixture models in the broader Mplus-style structural equation framework. Lubke and Muthen [-@lubkemuthen2005] conducted the first systematic Monte Carlo evaluation of factor mixture models and clarified the conditions under which the discrete-class component was empirically distinguishable from continuous heterogeneity. Clark and colleagues [-@clark2013] surveyed the modelling strategies and presented an applied analysis of psychiatric comorbidity.

Growth mixture models and group-based trajectory models extend the finite-mixture framework to longitudinal outcome trajectories.

- GMMs posit class-specific trajectories with within-class random-effects variation; group-based models constrain within-class variance to zero.
- Nagin's group-based approach originated in criminology and is now widely used in epidemiology and clinical research.

- Both families are the longitudinal analogues of the latent-class DGP developed in the present paper.

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In parallel, the longitudinal extension of these ideas yielded the growth mixture model [@muthenshedden1999; @muthen2001; @muthen2004] and the closely related group-based trajectory model [@nagin1999; @nagin2005]. Both posit that a longitudinal outcome trajectory is generated by a finite mixture of class-specific trajectories. Growth mixture models permit within-class random-effects variation around the class-specific mean trajectory; group-based trajectory models constrain within-class variance to zero, so each class has a single deterministic trajectory. Nagin's group-based approach was developed primarily in criminology to characterise developmental trajectories of antisocial behaviour, and has been widely applied in epidemiology, public health, and clinical research where trajectory heterogeneity is the principal scientific question.

Overextraction hazards tempered early optimism: GMMs can spuriously identify classes from non-gaussian single-component data, and reliable class recovery requires large samples.

- Bauer and Curran showed GMMs extract apparent classes from non-gaussian single-component DGPs.
- Lubke and Muthen documented analogous FMM hazards and identified joint conditions on N, class separation, and parameter differences.
- Nylund et al. systematised class enumeration via Monte Carlo evaluation of information criteria and BLRT; the cumulative message is that hundreds to thousands of participants are required.

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The methodological optimism around growth and factor mixture models during the early 2000s was tempered by a series of papers documenting overextraction hazards. Bauer and Curran [-@bauercurran2003], in Psychological Methods, demonstrated through simulation that growth mixture models fitted to data generated from a single non-gaussian distribution can spuriously identify multiple latent classes; the apparent classes are artefacts of distributional non-normality rather than evidence of substantive subpopulation structure. Bauer [-@bauer2007] elaborated on the implications for substantive *psychological research and recommended caution in interpreting class structure without independent validation. Lubke and Muthen [-@lubkemuthen2005] documented the analogous concerns for factor mixture models, identifying the joint conditions on sample size, class separation, and class-specific parameter differences under which class structure is recoverable. Nylund and colleagues [-@nylund2007] systematised the class-enumeration question through Monte Carlo evaluation of information criteria and likelihood-ratio bootstrap procedures. The cumulative message of this literature is that mixture models can extract apparent class structure where none exists, and that distinguishing genuine from spurious class structure requires either substantial samples (typically several hundred to several thousand participants) or strong prior information.*

2.6 Translation into longitudinal biostatistics

Verbeke and Lesaffre introduced finite-mixture random-effects distributions into biostatistics via longitudinal HIV data, with Zhang and Davidian extending to semi-parametric specifications.

- The model augments a standard random-intercept-and-slope LMM with a discrete latent class indicator on the random-effects population.
- Their motivating application was qualitatively distinct CD4 trajectories in HIV-infected patient subgroups.
- Zhang and Davidian allowed more flexible heterogeneity distributions via semi-parametric mixture specifications.

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Mixture and latent-class methods entered biostatistical practice through several complementary channels. Verbeke and Lesaffre [-@verbekelesaffre1996], in the Journal of the American Statistical Association, proposed a linear mixed-effects model in which the random-effects population was itself a finite mixture of multivariate normals. Their motivating application was longitudinal HIV-infection data in which subgroups of patients displayed qualitatively different CD4 trajectories. The model is essentially a growth mixture model in biostatistical clothing, with the standard random-intercept-and-slope linear mixed model augmented by a discrete latent class indicator. Zhang and Davidian [-@zhangdavidian2001] extended the framework to allow more flexible random-effects distributions via semi-parametric mixture specifications.

The `lcmm` and `flexmix` R packages translate finite-mixture longitudinal models into accessible software for biostatistical applications.

- `lcmm` implements joint latent-class mixed models for longitudinal outcomes with optional joint survival modelling.
- `flexmix` provides a general regression-mixture framework with user-definable component models including mixed-effects specifications.

- Both have been used in pharmacokinetic, cognitive ageing, and clinical subgroup discovery applications.

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Proust-Lima and colleagues developed a more comprehensive software framework with the `lcmm` R package [@proustlima2017], which implements joint latent-class mixed models for longitudinal continuous and ordinal outcomes, with optional joint modelling of survival outcomes via shared latent class structure. The `lcmm` framework has been deployed in cognitive ageing research, Alzheimer's disease progression modelling, and clinical-trial subgroup discovery. The `flexmix` package in R [@leisch2004; @grunleisch2008] provides a more general infrastructure for fitting mixtures of regression-style models with user-supplied component specifications, and has been used in pharmacokinetic and pharmacodynamic modelling where individual-level variation in dose-response curves motivates a mixture structure.

Mixture and latent-class methods have not been systematically applied to aggregated N-of-1 designs; the biomarker-moderation question has been encoded exclusively as a continuous interaction.

- Published N-of-1 methods focus on within-participant estimation, Bayesian hierarchical models, and meta-analytic aggregation.
- Biomarker-moderation power calculations use continuous interaction terms with no parallel latent-class formulation.
- The mixture-formulation question appears unaddressed in the existing N-of-1 methodology literature.

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Despite this translation activity, mixture and latent-class methods have not, to the best of our knowledge, been systematically applied to aggregated N-of-1 trial designs. The published N-of-1 methods literature is dominated by within-participant effect estimation [@zucker1997; @lillie2011], Bayesian hierarchical models for participant-specific effects [@zucker2010], and aggregated-evidence meta-analytic frameworks [@schork2015; @senn2018]. The biomarker-moderation question has received occasional treatment under the heading of personalised-medicine power calculation [@gabler2011; @hendrickson2020], but in each case the moderation has been encoded as a continuous interaction term in a fixed-effects analysis, with no parallel *treatment of the latent-class formulation. The mixture-formulation question appears to be unaddressed in the existing N-of-1 methodology literature.*

2.7 Why this matters for N-of-1 designs specifically

Three features of the aggregated N-of-1 design make the continuous-versus-discrete moderation question particularly consequential for design and analysis decisions.

Under a latent-class DGP, class membership is fixed across drug states and is in principle recoverable from the full trajectory even when on-drug contrasts are noisy.

- Continuous-heterogeneity DGPs encode moderation in how the within-participant contrast varies with the biomarker.
- Latent-class DGPs encode moderation in class membership, which is invariant to drug state.
- The two encodings imply different efficient analysis strategies and different power profiles under carryover.

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First, the within-participant longitudinal structure of an N-of-1 trial provides repeated measurements at multiple drug states (on-drug, off-drug, in carryover). Under a continuous-heterogeneity DGP, the moderation signal is encoded in how the within-participant treatment contrast varies with the biomarker. Under a latent-class DGP, the moderation signal is encoded in class membership, which is fixed across all measurements within a participant and is therefore in principle recoverable from the full trajectory even when individual on-drug contrasts are noisy. The two encodings imply different efficient analysis strategies and different power profiles under carryover [@hendrickson2020].

N-of-1 trial sample sizes are well below the threshold at which mixture-model identifiability has been validated, so a misspecified single-component analysis may be the more reliable tool.

- Psychometric identifiability validation requires hundreds to thousands of participants; N-of-1 trials operate at $N \in [30, 150]$.
- Even a biologically correct latent-class DGP may yield underpowered or unidentified mixture estimates at trial-relevant N.
- The mixture-versus-mitigation trade-off is an empirical question not addressed at N-of-1-relevant sample sizes.

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Second, N-of-1 trials are typically conducted at sample sizes of $N \in [30, 150]$, well below the sample sizes at which mixture-model identifiability has been validated in the psychometric literature. The identifiability cautions documented by @bauercurran2003 and @lubkemuthen2005 therefore apply with full force, and the analysis-strategy implications are not trivial: even if a latent-class DGP is the biologically correct generative model, mixture analysis at trial-relevant sample sizes may be either underpowered or unidentified, and a misspecified single-component analysis may be the more reliable inferential tool despite its bias. Resolving this trade-off is in part an empirical question that the existing psychometric simulation literature has not addressed at N-of-1-relevant sample sizes.

Carryover erodes the moderation signal directly under continuous-heterogeneity but only indirectly under latent-class, predicting divergent power-loss profiles.

- Under continuous heterogeneity, carryover attenuates D_{it} and thereby directly attenuates the moderation signal.
- Under latent-class structure, carryover erodes within-class drug contrasts but does not directly attenuate class membership.
- The two formulations predict different power-loss profiles; this difference is empirically large in the companion carryover- sensitivity work.

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Third, carryover, which enters the analysis as a continuous decay in the effective drug indicator $D_{it} = \exp(-\lambda t_{sd,it})$, interacts differently with the two moderation encodings. In the continuous-heterogeneity formulation, carryover erodes the moderation signal directly because the moderation enters as a product with D_{it}. In the latent-class formulation, carryover erodes the within-class drug contrast but does not directly attenuate class membership; if class membership can be recovered from the full trajectory, some moderation signal can survive carryover that would be lost under the continuous-heterogeneity analysis. The two formulations therefore predict different power-loss profiles under carryover, and this difference is empirically large in the simulation literature accumulated by the companion carryover-sensitivity work.

2.8 Aims and structure of the paper

The paper has four aims: formalise latent-class DGPs for N-of-1 trials, characterise estimation strategies at small N, benchmark power via sim-

ulation, and re-analyse the prazosin-PTSD dataset.

- Aims 1–2 are methodological: formal specification and estimation/diagnostic characterisation with explicit identifiability conditions.
- Aim 3 is computational: a factorial simulation benchmarking mixture-DGP against continuous-DGP power under carryover.
- Aim 4 is applied: re-analysis of the @hendrickson2020 prazosin-PTSD data under the latent-class formulation.

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This paper has four aims. The first is to formalise the latent-class and finite-mixture formulations of biomarker-treatment interaction in aggregated N-of-1 trials, with explicit identifiability conditions specific to the trial-relevant sample-size regime. The second is to characterise estimation strategies (expectation-maximisation, Bayesian samplers, candidate R packages) and the diagnostics required to distinguish genuine from spurious class structure at small N. The third is to report a factorial simulation study that benchmarks mixture-DGP and continuous-DGP power against the standard linear-mixed-model analysis and a class-aware analysis under realistic carryover. The fourth is to apply the developed methods to the prazosin-PTSD biomarker dataset of @hendrickson2020 and to compare conclusions with those obtained under the continuous-moderation analysis reported in the original paper.

The paper moves from notation and DGP specification through MVN approximation, identifiability analysis, model taxonomy, and spectrum organisation to analysis implications and a discussion of open questions.

- Sections on the faithful DGP, MVN approximation, and failure regimes develop the theoretical core.
- The finite-mixture taxonomy and biomarker-moderation spectrum organise the model landscape systematically.
- The discussion addresses three open questions and situates the companion five-study simulation programme.

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The remainder of the paper is organised as follows. We begin in the notation section with a glossary that fixes terminology used throughout. The latent-class generative form, its MVN second-moment approximation, and *the regimes in which the approximation breaks down are developed in three subsequent sections. The implications for power under carryover and the identifiability constraints that bound mixture analysis at trial-relevant sample sizes are addressed next. A finite-mixture taxonomy section locates the relevant model families (LCA, GMM, FMM, regression mixtures); a biomarker-moderation spectrum section organises the available DGP encodings moment by moment. Implications for analysis and design, and software for mixture analyses, are then drawn together. The discussion section addresses three open methodological questions and references the five-study simulation programme described in the companion simulation plan, which constitutes the empirical core of the present line of work and is the next deliverable in the programme.*

3 Notation

Core notation: B_i is the biomarker, BR_{it} the biological-response component, D_{it} the continuous-decay drug indicator, and Z_i the unobserved class label.

- Outcome is decomposed into BR_{it}, TV_{it}, and PB_{it} following the Hendrickson framework.
- $D_{it} = \exp(-\lambda t_{sd,it})$ off-drug and 1 on-drug; $\lambda = \ln 2/t_{1/2}$.
- The DGP and analysis model are permitted to use different carryover half-lives.

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Throughout, B_i denotes the pre-treatment biomarker value for participant i. BR_{it} denotes the biological-response component of outcome for participant i at time t, decomposed in the Hendrickson framework into three latent response factors: BR_{it} itself, a time-variant component TV_{it}, and a placebo-belief component PB_{it}. D_{it} denotes drug state at time t in the analysis-side continuous-decay form $D_{it} = \exp(-\lambda t_{sd,it})$ during off-drug periods and $D_{it} = 1$ during on-drug periods, with $t_{sd,it}$ the time since drug discontinuation and λ the decay rate. Z_i denotes an unobserved class label. The half-life of carryover is $t_{1/2} = \ln 2/\lambda$ and is allowed to differ between the DGP and the analysis model.

The principal psychometric and econometric terms used below are:

- **Latent class.** An unobserved categorical subgroup, equivalent in trial vocabulary to an unobserved responder/non-responder stratum.
- **Latent factor.** An unobserved continuous trait governing covariation among observed variables, equivalent to a frailty or to a random intercept in a linear mixed model.

- **Class-conditional distribution f_z.** The outcome distribution within a single subgroup.
- **Mixing proportion π.** The prevalence of a given subgroup in the population.
- **Measurement invariance.** Whether a model applies identically across subpopulations.
- **Growth mixture model (GMM).** A longitudinal mixed model in which the population of random effects is itself a finite mixture.
- **Factor mixture model (FMM).** A hybrid model with discrete classes at the upper level and within-class continuous heterogeneity at the lower level.
- **Regression mixture / mixture of experts.** A finite mixture in which the gating function (class-membership probability) depends on covariates and within-class regressions are class-specific.
- **EM algorithm.** Expectation-maximisation applied to recover unobserved class labels jointly with class-conditional model parameters.

4 Faithful latent-class data-generating processes

If a candidate predictive biomarker indexes an unobserved responder/non-responder dichotomy, the mechanistically faithful data-generating process is a finite mixture:

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{it}),$$

where Z_i is the latent class indicator, $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ is the class-membership probability (typically a logistic function of the biomarker), and the class-conditional response distributions f_0 and f_1 differ in their drug-response parameters: peak response, onset rate, or both. Within-class temporal structure follows the same modGompertz form used in single-class formulations, with class-specific parameters substituted.

> **The faithful latent-class DGP is a finite mixture with biomarker-dependent class-membership probability and class-specific modGompertz drug-response distributions.**
>
> - $Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i))$ with $\pi(B_i)$ typically a logistic function of the biomarker.
> - Class-conditional distributions f_0 and f_1 differ in drug-response parameters (peak response, onset rate, or both).
> - Within-class temporal structure uses the same modGompertz form as single-class formulations.

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If a candidate predictive biomarker indexes an unobserved responder/non-responder dichotomy, the mechanistically faithful data-generating process is a finite mixture, with $Z_i \mid B_i$ following a Bernoulli distribution with class-membership probability $\pi(B_i)$ (typically logistic in the biomarker), and the class-conditional response distributions f_0 and f_1 differing in their drug-response parameters. Within-class temporal structure uses the same mod-Gompertz form as single-class formulations with class-specific parameters substituted.

Class-label invariance preserves moderation information across carryover that the MVN-covariance encoding, where the signal is entirely in the second moment, cannot retain.

- Z_i is fixed across drug states; it carries participant-level moderation information even when between-class mean separation is attenuated.
- Under full carryover washout, class-conditional means converge but Z_i remains recoverable from the full trajectory.
- Under MVN-covariance encoding the carryover erosion of the covariance is direct and no residual signal survives washout.

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We note two consequences of this DGP form that are central to what follows. First, the class label Z_i is fixed across all measurements within a participant; it is invariant to drug state and to time. Second, the class-conditional mean trajectory does depend on D_{it}: under full carryover washout ($D_{it} \to 0$) the two class-conditional means converge if the responder/non-responder distinction is purely in drug response. Class-label invariance therefore preserves participant-level information about the moderation signal that an analysis with access to Z_i could exploit, even where instantaneous between-class mean separation in BR is fully attenuated. The MVN-covariance encoding of biomarker moderation, in contrast, has no such latent-label information to exploit: the covariance is the entire signal, and its erosion under carryover is direct.

5 MVN approximation to mixture-induced moderation

A two-component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\mathrm{Cor}(B, BR) = c_{bm}$. Under within-class independence between B_i and BR_{it},

$$c_{bm}^2 = \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\mathrm{Var}(B)} \cdot \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\mathrm{Var}(BR)},$$

so that c_{bm}^2 is the product of the between-class variance fractions in B and in BR. The marginal correlation that an MVN can reproduce is therefore bounded by the geometric mean of the two intra-class correlations; if either class separation is small, the achievable marginal c_{bm} is small even when class structure is otherwise well-defined.

The achievable marginal c_{bm} under an MVN approximation is bounded by the geometric mean of the two between-class variance fractions.

- c_{bm}^2 equals the product of the between-class variance fractions in B and in BR.
- Small class separation in either variable limits the achievable marginal c_{bm} even when class structure is otherwise well-defined.
- The MVN approximation reproduces the mixture's first two moments under within-class independence.

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A two-component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\mathrm{Cor}(B, BR) = c_{bm}$. The achievable marginal c_{bm} is bounded by the geometric mean of the two between-class variance fractions; if either class separation is small, the achievable c_{bm} is small even when class structure is otherwise well-defined.

Within the regularity conditions, the MVN second-moment approximation is adequate for LMM power calculations and captures differential correlation across drug states.

- The approximation yields elevated correlation on-drug and shrinkage toward zero off-drug after carryover washout.
- Adequacy requires adequate class overlap and π bounded away from 0 and 1.
- For t-tests on $\beta_{bm:D}$ this approximation captures covariance consequences without committing to a discrete DGP.

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Under mild regularity conditions (adequate class overlap and mixing proportions π bounded away from 0 and 1), the mixture's second-moment structure is well-approximated by a single MVN with differential correlation across drug states: elevated correlation where class-conditional means separate (on-drug) and shrinkage towards zero where they coincide (off-drug after carryover washout). For power calculations driven by t-tests on $\beta_{bm:D}$ in a linear mixed-effects model, this approximation is adequate within the regularity conditions and captures the covariance consequences of latent-class structure without committing to a discrete generative form.

6 Failure regimes of the MVN approximation

The MVN approximation is misleading in three identifiable regimes, each with a distinct empirical signature.

- **Bimodal responses.** When f_0 and f_1 are well-separated (non-responders show near-zero drug effect and responders show a large effect), the marginal distribution of BR is bimodal: a histogram of on-drug response would show two peaks rather than a single peak with gaussian scatter around it. No MVN matches the tails of a bimodal distribution, and power under the true mixture DGP will exceed the MVN-LMM prediction because a well-specified mixture analysis can exploit the bimodality. The pattern occurs in clinical settings where a subpopulation genuinely does not respond to a treatment, as in certain targeted oncology indications.
- **Strong class-membership gradient.** When $\pi(B_i)$ is steep, approaching a step function at a biomarker threshold, the biomarker behaves nearly deterministically as a class label. A thresholded mean-moderation specification (a thresholded Architecture-A variant) then becomes the better approximation, because the biomarker-response relationship is dominated by its mean-structure component. The corresponding clinical scenario is a diagnostic biomarker with near-perfect sensitivity and specificity for the responsive phenotype.
- **Class-varying covariance.** If autocorrelation, residual variance, or placebo-response parameters differ between classes, no single-component MVN can represent the generative structure. Analysis models assuming constant covariance are misspecified regardless of whether they encode the biomarker interaction in the mean or in the covariance. This is the latent-class analogue of heteroscedastic residual variance addressed in standard mixed- effects practice with `varIdent` or `weights` specifications, except that the stratifying variable is unobserved.

7 Implications for power under carryover

The power gap between a second-moment analysis and a mixture-faithful analysis upper-bounds what a class-aware procedure can recover, but the

bound is asymptotic and may reverse at small N.

- The 40–60% covariance-only power loss reflects direct erosion of a second-moment signal; class membership is carryover-invariant.
- The gap between covariance-only and mixture-faithful power estimates upper-bounds the recoverable power.
- At finite N in the trial-relevant range, finite-sample bias can drive the realised gap below the bound or reverse its sign.

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The substantial covariance-only power loss under carryover (40–60% across designs in @hendrickson2020) reflects erosion of a second-moment signal. A mixture-model analysis fitted to mixture-DGP data can in principle recover additional power by estimating class membership directly, because the class label is carryover-invariant. The gap between the covariance-only carryover-sensitive power estimate and the mixture-faithful power estimate upper-bounds what a mixture analysis could recover; whether this bound is tight in the trial-relevant parameter regime is an empirical question not addressed by existing simulation infrastructure. The bound is asymptotic: at finite N, finite-sample bias can drive the realised gap below the bound or, in adverse identifiability regimes, reverse its sign.

The Study 1 factorial crosses two DGPs and two analyses to decompose approximation error from analysis-model mis-specification and to verify Type I error control.

- The mixture-DGP/MVN-analysis vs. mixture-DGP/mixture-analysis contrast quantifies power recoverable by a faithful analysis.
- The two MVN-analysis cells quantify approximation error in the second-moment encoding.
- A null check on the MVN-DGP/mixture-analysis cell verifies no spurious power extraction.

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A focused factorial simulation isolating second-moment approximation error from analysis-model mis-specification is described in the companion simulation plan (Study 1: mixture-vs-MVN factorial). The factorial crosses two DGPs (mixture and MVN) with two analyses (linear mixed model and joint *latent-class mixed model). The mixture-DGP/MVN-analysis versus mixture-DGP/mixture-analysis contrast quantifies power recoverable by a faithful analysis model; the difference between the two MVN-analysis cells quantifies approximation error. A null check on the MVN-DGP/mixture-analysis cell verifies the mixture model does not extract spurious power on data with no latent class structure.*

8 Identifiability at trial-relevant sample sizes

Mixture EM requires either informative class-probability priors or strong class separation; neither is assured at N-of-1 sample sizes, which fall well below identifiability validation thresholds.

- Posterior entropy stabilisation requires hundreds to thousands of participants in published psychometric applications.
- N-of-1 trials operate at $N \in [30, 150]$, well below this range.
- The identifiability cost is acute precisely when the moderation effect's existence is uncertain.

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Mixture modelling is attractive on faithfulness grounds but carries identifiability costs that are acute in N-of-1 settings. The EM algorithm for finite mixtures of mixed-effects models requires either informative priors on class-membership probabilities or strong separation between class-conditional response curves; neither is assured in a trial designed to detect a moderation effect whose existence is itself uncertain. The standard diagnostic for mixture identifiability, the entropy of posterior class-membership probabilities, requires substantial sample sizes (N several hundred to several thousand) to stabilise at values that support confident class extraction in published psychometric applications [@bauercurran2003; @bauer2007; @nylund2007]. Aggregated N-of-1 trials of the size contemplated in current biomarker applications, $N \in [30, 150]$, *are well below this range.*

The operative question is not faithfulness but whether the fidelity gain is recoverable at realistic N, or whether standard analysis mitigations offer a better power-per-complexity trade.

- Mixture analysis may be underpowered or unidentified even under a genuinely mixture-structured DGP at trial-relevant N.
- Analysis-strategy mitigations within standard mixed-effects practice have better-understood inferential properties at small N.
- Study 5 of the companion simulation plan addresses this directly with small-N identifiability and Type I error checks.

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The methodological implication is that mixture analysis at trial-relevant sample sizes may be either underpowered or unidentified, even when the underlying DGP is genuinely mixture-structured. The question is therefore not whether mixture formulations are more faithful, but whether the fidelity gain is recoverable at realistic sample sizes, or whether analysis-strategy mitigations available within standard mixed-effects practice (restricted analysis, weighting, within-subject contrasts; see @hendrickson2020) deliver a better power-per-complexity trade. These mitigations have better-understood inferential properties at the relevant sample sizes. Resolving the mixture-versus-mitigation question empirically requires the simulation programme described in the companion simulation plan; Study 5 in particular addresses small-N identifiability and type I error explicitly.

9 Finite-mixture taxonomy

Four families of finite mixture model from the psychometric and econometric literatures bear directly on the biomarker-treatment interaction problem. In this section we shall introduce each family, its origin, and its specific relevance to the N-of-1 trial setting.

9.1 Latent class and latent profile analysis

LCA and LPA are cross-sectional mixture models estimating class membership from observed indicators; in the trial context they characterise responder classes from response trajectories.

- LCA uses categorical indicators; LPA uses continuous indicators; both return posterior class-membership probabilities.
- In the biomarker-trial context, an LPA-style model estimates responder-class probabilities from on-drug and off-drug response trajectories.
- Classical estimation is via EM; standard R implementations include `poLCA` and Latent GOLD.

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Latent class analysis and latent profile analysis are mixture models for cross-sectional data. Each participant belongs to one of a small number of un-

observed classes; the observed variables have class-specific marginal distributions. LCA uses categorical indicators; LPA uses continuous indicators. A mixture fitted to baseline biomarkers alone returns posterior class-membership probabilities that can be used as covariates in subsequent outcome analysis, the two-stage analytic strategy standard in psychometric applications. In the biomarker-trial context, an LPA-style model posits two or more responder classes with class-specific marginal distributions of on-drug and off-drug response, and estimates class-membership probabilities from the observed response trajectories rather than from baseline covariates alone. Classical references include @lazarsfeldhenry1968, @goodman1974, @clogg1995, and @mclachlanpeel2000; the Vermunt and Magidson Latent GOLD software and the `poLCA` R package implement the standard EM estimation routines.

9.2 Growth mixture models and group-based trajectory models

GMMs extend LPA to longitudinal data with class-specific drug-response curves and within-class random-effects variation; group-based trajectory models constrain within-class variance to zero.

- A GMM is a longitudinal LMM with the random-effects population replaced by a mixture of multivariate normals.
- Group-based trajectory models are the degenerate case with zero within-class random-effect variance.
- Both families model longitudinal trajectory heterogeneity as the principal scientific question.

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Growth mixture models [@muthenshedden1999; @muthen2001; @muthen2004] extend LPA to longitudinal data by specifying class-specific growth curves (or, in the trial setting, drug-response curves) with within-class random-effects variation. A GMM is a longitudinal linear mixed model in which the population distribution of random effects is a mixture of multivariate normals, each corresponding to a distinct trajectory class. Group-based trajectory models [@nagin1999; @nagin2005] are the degenerate case in which within-class random-effect variances are set to zero, so that participants within a class share an identical trajectory up to observation-level residual noise.

GMM is the closest analogue of the faithful DGP in Section 3, but Bauer and Curran's overextraction cautions apply with full force at N-of-1 sample sizes.

- A biomarker-gated GMM is essentially the faithful latent-class DGP with drug-state-invariant class labels.
- GMM can extract apparent class structure from single-class non-gaussian data at realistic sample sizes.

- N-of-1 participant counts are typically well below those at which GMM class-count recovery has been validated.

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GMM is the closest psychometric analogue of the latent-class generative form developed above: each participant belongs to a latent class with its own drug-response trajectory, and the class label is drug-state-invariant. A GMM that uses the biomarker as a class-membership predictor is essentially the faithful generative model written down in Section 3 of this paper. Bauer and Curran [-@bauercurran2003] and Bauer [-@bauer2007] document the identifiability and overextraction hazards that accompany GMM estimation at realistic sample sizes, showing in particular that GMM can extract apparent class structure from data generated by single-class non-gaussian processes. These cautions apply with full force to aggregated N-of-1 trials, where participant counts are typically well below the sample sizes at which GMM class-count recovery has been validated.

9.3 Factor mixture models

Factor mixture models combine discrete class structure with within-class continuous factor heterogeneity, subsuming pure factor analysis and pure LCA as limiting cases.

- Within-class variation is governed by a factor model; between-class variation by class-specific means, loadings, or residual covariances.
- The pure continuous (single-class factor) and pure discrete (LCA) cases are special cases of the FMM framework.
- FMMs match a wider range of biological scenarios at the cost of higher estimation requirements.

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Factor mixture models [@yung1997; @arminger1999; @lubkemuthen2005] combine continuous latent factors with discrete latent classes, yielding a generative structure in which within-class variation is governed by a factor model and between-class variation is governed by class-specific means and, optionally, class-specific factor loadings or residual covariances. The FMM framework subsumes both the pure continuous case (single-class factor model) and the pure discrete case (LCA with no within-class structure) as limits.

In FMM, the biomarker can function simultaneously as a noisy class indicator via the gating function and a continuous within-class modulator via factor loadings.

- The upper level assigns participants to classes as in LCA; the lower level gives each class its own random-effects structure.
- The biological hypothesis is distinct responder subtypes with additional continuous heterogeneity within each subtype.
- The hybrid representation is more biologically flexible than either pure LCA or pure factor analysis.

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FMMs are two-level hierarchies. At the upper level, participants are assigned to one of a small number of classes, as in LCA. At the lower level, each class has its own within-class random-effects structure. The biological hypothesis the FMM encodes is that there are genuinely distinct responder subtypes and that, within subtype, additional continuous heterogeneity is driven by covariates or unmeasured individual differences. For the biomarker-moderation problem, FMM provides a principled way to represent a biomarker as both a noisy class indicator (via the gating function) and a continuous modulator of within-class response (via factor loadings on the response factor). The hybrid representation matches more biological scenarios than either pure LCA or pure factor analysis, at the cost of correspondingly higher estimation requirements.

9.4 Regression mixtures and mixtures of experts

Regression mixtures replace a single regression surface with class-specific surfaces governed by a covariate-dependent gating function, paralleling propensity-score stratification in causal inference.

- Each class has its own regression surface; the gating model governs which surface applies to each participant.
- The biomarker enters both the gating function and (via class membership) the within-class regression coefficient on treatment.
- The two-stage structure parallels propensity-score stratification followed by within-stratum effect estimation.

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Regression mixtures [@desarbocron1988; @wedeldesarbo1995; @jedidi1997] and the closely related mixture-of-experts architecture [@jacobs1991] allow class-specific regression coefficients with class-membership probabilities (the gating function) that depend on covariates. An ordinary linear model with a treatment-by-biomarker interaction posits a single regression surface in which the treatment slope varies smoothly with the biomarker. A regression mixture instead posits two or more distinct regression surfaces, one per class, with a separate covariate-dependent model governing which surface applies to which participant. The two-stage structure (classification by the gating model, then prediction by the within-class regression) parallels the two-stage structure of causal-inference methods that use propensity scores to stratify before estimating within-stratum treatment effects.

In the trial simulation idiom, regression mixtures implement a DGP where the biomarker governs both responder probability and within-responder effect magnitude; `flexmix` provides the R implementation.

- The biomarker enters the gating function and determines within-responder drug effect magnitude.
- `flexmix` implements this family with user-definable component models including mixed-effects specifications.
- This family is suitable for repeated-measures DGP construction in aggregated N-of-1 simulation.

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In the trial-simulation idiom, regression mixtures correspond to a DGP in which the biomarker governs both the probability of being a responder and the magnitude of the within-responder drug effect. The `flexmix` package in R [@leisch2004; @grunleisch2008] implements this family with user-definable component models, including mixed-effects components suitable for repeated-measures data.

10 The biomarker-moderation spectrum

Locating the covariance-only formulation within the broader DGP spectrum clarifies which features of the carryover-induced power loss are properties of the specification rather than the science.

- The spectrum is organised by which moments the biomarker moderates: mean, covariance, both, or higher moments.
- Covariance-only is the Hendrickson formulation; mean-and- covariance and higher-moment forms are the generalisations.

- Understanding the spectrum is prerequisite to a complete sensitivity analysis of carryover-induced power loss.

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The covariance-only formulation of @hendrickson2020 specifies a biomarker-dependent covariance structure with population means held constant across biomarker values. Locating this formulation within the broader spectrum of biomarker-moderated DGPs is useful because the moment-by-moment organisation of the spectrum (biomarker moderating the mean, the covariance, both, or higher moments) clarifies which features of the carryover-induced power loss are properties of the specification itself rather than of the underlying scientific question.

10.1 Covariance-only moderation

The covariance-only specification fixes the conditional mean and encodes the interaction entirely in how strongly the biomarker co-varies with response during on-drug versus off-drug periods.

- High- and low-biomarker participants share the same expected drug response but differ in their biomarker-response covariance.
- The interaction signal resides entirely in the second moment and is eroded directly by carryover.
- This is the core DGP in @hendrickson2020.

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The covariance-only specification holds $E[BR_{it}]$ fixed across biomarker strata and encodes the interaction entirely in $\mathrm{Cov}(B_i, BR_{it} \mid D_{it})$. High- and low-biomarker participants have the same expected drug response at any given time but differ in the covariance between biomarker and response, specifically in how strongly the biomarker co-varies with response during on-drug versus off-drug periods.

Covariance-only is an unusual restriction in the psychometric tradition; its tractability for MVN power simulation is its main appeal, but it lacks a clean natural generative mechanism.

- FMM and GMM parameterisations typically allow class-specific means at minimum; the means-constant-covariance-varies case tests measurement invariance, not primary DGP structure.

- The encoding's appeal is tractability: the entire DGP reduces to a single MVN draw with a constructed covariance.
- Its cost is the absence of a natural mechanistic interpretation.

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This is an unusual restriction within the psychometric mixture tradition. Most standard FMM and GMM parameterisations permit class-specific means at a minimum, and the means-constant-covariance- varies case is typically encountered only as a constrained submodel used for testing measurement-invariance hypotheses [@meredith1993; @widamanreise1997]. The corresponding biostatistical model has two subgroups with identical expected outcomes but differing residual variance or within-subject autocorrelation; such models are occasionally fitted as sensitivity analyses but rarely assumed as the primary DGP. The covariance-only encoding's appeal lies in its tractability for power simulation under multivariate normality: the entire DGP reduces to drawing from a single MVN with a constructed covariance. Its cost is that it does not correspond cleanly to any natural generative mechanism.

10.2 Combined mean and covariance moderation

The Arminger mean-and-covariance form is the general psychometric parameterisation, subsuming both pure mean moderation and covariance-only as special cases.

- Both the expected-response vector and the residual covariance matrix may differ between classes and depend on covariates.
- Mean-moderation only and covariance-only are limiting cases.
- This is the general form against which both Architectures A and B are constrained submodels.

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Arminger and colleagues [-@arminger1999] develop finite mixtures of conditional mean- and covariance-structure models in which both the expected-response vector and the residual covariance matrix differ between classes and may depend on observed covariates within class. This is the general psycho-

metric form: it reduces to a mean-moderation model in the limit of a single class with covariate-dependent mean only, and to the covariance-only encoding in the limit of class-invariant means with covariate-dependent covariance only.

Under combined moderation, the mean channel is robust to carryover and the covariance channel is eroded, predicting intermediate power loss and making this the most biologically plausible regime.

- Both channels carry class-membership information; the mean channel via $E[BR \mid B]$, the covariance channel via $\mathrm{Var}[BR \mid B]$.
- Carryover attenuates the covariance channel but not the mean channel, yielding intermediate power loss.
- The combined specification is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials.

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In the combined form, responders and non-responders differ both in typical drug response (mean moderation) and in how noisily that response is expressed within class (covariance moderation). Both channels carry information about class membership: the mean channel via $E[BR \mid B]$, the covariance channel via $\mathrm{Var}[BR \mid B]$. Under carryover, the mean channel is comparatively robust because class-conditional means need not converge off-drug; the covariance channel is eroded by construction because off-drug covariance is attenuated. A combined-specification simulation should therefore exhibit power loss intermediate between the pure-mean and pure-covariance extremes; this is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials. Verbeke and Lesaffre [-@verbekelesaffre1996] provide the analogous treatment for linear mixed-effects models with heterogeneity in the random-effects population, which is the more direct generalisation of the repeated-measures structure used in N-of-1 trial simulation.

10.3 Location-scale moderation

GAMLSS-style distributional regression allows the biomarker to enter the mean, variance, and higher distributional moments simultaneously via separate link functions.

- Standard regression models only the conditional mean; GAMLSS models the mean, variance, skewness, and kurtosis each as a separate function of covariates.
- The familiar special case is heteroscedastic linear regression with covariate-dependent residual variance.
- GAMLSS generalises this to arbitrary distributional shape.

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Generalised additive models for location, scale, and shape [@gamlss2005] and the related distributional regression tradition [@klein2015] allow the biomarker to enter multiple parameters of the outcome distribution simultaneously through separate link functions. In plain terms, a standard regression models the conditional mean of an outcome as a function of covariates; a GAMLSS-style distributional regression models the conditional mean, the conditional variance, and if appropriate the conditional skewness and kurtosis, each as a separate function of covariates. The familiar special case is heteroscedastic linear regression with covariate-dependent residual variance; GAMLSS generalises this to arbitrary distributional shape.

GAMLSS occupies an intermediate position between Architecture A and B, allowing joint mean and scale moderation without discrete class structure.

- A GAMLSS DGP allows a shifted mean (Architecture A location effect) and altered variability (Architecture B analogue) jointly.
- It is the continuous-heterogeneity analogue of FMM, combining mean and scale effects without requiring class structure.
- The intermediate position is attractive for sensitivity analysis as it avoids committing to a class count.

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In the biomarker-trial context, a GAMLSS-based DGP would allow high-biomarker participants to have both a shifted mean drug response (a location effect, which is Architecture A) and altered response variability (a scale effect, which is analogous to Architecture B's covariance moderation but operating on marginal rather than joint moments), without requiring discrete class structure. The GAMLSS framework is thus the continuous-heterogeneity analogue of FMM and occupies an intermediate position between pure Architecture A (mean only) and pure Architecture B (covariance only). This intermediate position is attractive for sensitivity analysis because it does not require committing to a specific number of latent classes.

10.4 Heterogeneous random-effects models

Heterogeneous random-effects models replace the single normal random-slope distribution with a mixture, so responder and non-responder classes have distinct treatment-slope distributions.

- In the standard LMM the random treatment slope is drawn from a single normal; the heterogeneous extension replaces it with a finite mixture.
- Responders draw from a large-slope component; non-responders from a near-zero-slope component.
- The biomarker predicts which mixture component a participant comes from.

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Verbeke and Lesaffre [-@verbekelesaffre1996], Zhang and Davidian [-@zhangdavidian2001], and Proust-Lima and colleagues [-@proustlima2017] develop linear mixed-effects models in which the random-effects distribution is itself a finite mixture, allowing class-specific random-intercept and random-slope distributions. The biostatistical setting this extends is the standard linear mixed model with random intercept and random treatment slope, familiar from any longitudinal trial analysis using `nlme::lme` or `lme4::lmer`. In the standard model, the random treatment slope is drawn from a single normal distribution with mean equal to the average treatment effect and variance capturing individual heterogeneity. In the heterogeneous random-effects extension, that single distribution is replaced by a mixture: a fraction of participants are drawn from a 'responder' distribution with a large mean treatment slope and another fraction from a 'non-responder' distribution with a near-zero mean treatment slope. The biomarker predicts which mixture component a participant comes from.

`lcmm` is the closest available tooling for heterogeneous random-effects N-of-1 analyses, fitting a mixture on the random-effects distribution and returning posterior class probabilities.

- The framework represents responder heterogeneity at the participant-specific random treatment effect – the most clinically transparent representation.
- The biomarker predicts class membership rather than entering the fixed-effect mean structure.
- `lcmm` accepts standard LMM syntax and is the lowest-friction extension of the existing simulation framework.

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Applied to aggregated N-of-1 trial data, this framework represents responder heterogeneity directly at the participant-specific random treatment effect, which is arguably the most clinically transparent representation available: responders and non-responders have distinct random-slope distributions on the treatment indicator, and the biomarker predicts class membership rather than entering the analysis-model mean structure at the fixed-effect level. The `lcmm` package [@proustlima2017] implements this family and is the closest available tooling for the longitudinal aggregated N-of-1 use case: it accepts standard longitudinal mixed-model syntax, fits a mixture on the random-effects distribution, and returns posterior class probabilities for each participant. For an extension of an existing aggregated N-of-1 simulation framework, `lcmm` is the lowest-friction entry point.

11 Implications for analysis and design

Three implications follow from locating the covariance-only encoding within the broader spectrum.

The covariance-only DGP is the most restrictive case of a richer family; sensitivity analyses based on it alone risk overstating carryover-induced power loss.

- Covariance-only places the entire biomarker-outcome signal in a channel eroded by carryover by construction.
- Mean-structure signal survives carryover, attenuating overall power loss under combined or mean-only specifications.
- A complete sensitivity analysis should span pure mean moderation, pure covariance moderation, and a combined specification.

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First, the covariance-only parameterisation is the most restrictive operational case of a much richer family. The substantial power loss under carryover documented in @hendrickson2020 is a specific property of this restrictive case and need not generalise to the mean-plus-covariance or heterogeneous-random-effects variants that are standard in psychometric and biostatistical practice. A sensitivity analysis based solely on the covariance-only DGP risks overstating the carryover-induced power loss that a biomarker-guided aggregated N-of-1 trial should expect, because the covariance-only DGP places the entire biomarker-outcome signal in a channel that carryover erodes by construction. If a portion of the signal flows through the mean structure, as in pure mean moderation or in the combined specification of @arminger1999, that component survives carryover and overall power loss is correspondingly *attenuated. A complete sensitivity analysis should report power under at least three points on the spectrum: pure mean moderation, pure covariance moderation, and a combined specification.*

Identifiability concerns bound mixture-based analysis to a sensitivity and DGP role at N-of-1 sample sizes, not a primary analysis role.

- Published FMM applications involve hundreds to thousands of participants; N-of-1 trials at $N \in [30, 150]$ are an order of magnitude smaller.
- Mixture models should not be relied on as the primary analysis at trial-relevant N.
- Mixture analysis retains value as a sensitivity check and as a DGP stress-test for the primary linear-mixed analysis.

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Second, the identifiability concerns documented in the GMM and FMM literatures [@bauercurran2003; @bauer2007; @lubkemuthen2005; @nylund2007] bound what can be asked of mixture-based analysis models at N-of-1 trial sample sizes. Published FMM applications typically involve several hundred to several thousand participants; biomarker- trial applications with $N \in [30, 150]$ are an order of magnitude smaller. Analysis plans built around fitting `lcmm` or `flexmix` mixture models directly to trial data are unlikely to yield reliable class-membership inference at trial-relevant sample sizes and should not be relied on as the primary analysis. Mixture analysis nonetheless retains value as a sensitivity check (does class-structured residual pattern emerge from the pre-specified analysis?) and as a DGP for power simulation (does the primary analysis remain robust to class-structured departures from gaussianity?).

The heterogeneous random-slopes form is the natural DGP extension target: it admits latent-class generative structure while preserving `nlme::lme` analysis-model compatibility.

- Random-slopes parameterisation extends the DGP in a direction internal to the standard mixed-effects framework.
- The analysis model remains the pre-specified linear mixed-effects fit, preserving the audit chain.
- Direct comparison against the covariance-only baseline is possible under matched effect-size calibration.

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Third, if N-of-1 simulation infrastructure pursues a richer DGP in subsequent work, the natural target is the heterogeneous random-slopes form rather than a full FMM. The random-slopes parameterisation preserves the analysis model's compatibility with the existing `nlme::lme` machinery (the analysis model continues to be fitted as a single linear mixed model with participant-specific random treatment slopes) while admitting the biologically faithful latent-class generative structure at the DGP level. The simulation DGP is extended in a direction internal to the standard mixed-effects framework, and the analysis model remains the pre-specified linear mixed-effects fit; this preserves the audit chain of the existing framework and permits direct comparison against the covariance-only baseline under matched effect-size calibration.

12 Software for mixture analyses

R packages relevant to estimating the models surveyed above, in approximate order of fit to the longitudinal aggregated N-of-1 use case, are as follows.

- `lcmm`[3]. Joint latent-class mixed models for longitudinal continuous and ordinal outcomes; the closest available tooling for heterogeneous random-effects specification. Syntax is closely related to `nlme::lme` with a latent-class extension.
- `flexmix`[4,5]. General regression- mixture framework with user-definable component models; suitable for regression-mixture and mixture-of-experts specifications. More general than `lcmm` but requires more configuration.
- `mclust`[6]. Gaussian finite mixture models with a range of class-specific covariance parameterisations; supports[7] mean-plus-covariance specifications non-longitudinally. Useful for cross-sectional latent-profile analyses of baseline biomarker panels but does not natively handle repeated-measures structure.
- `gamlss`[8]. Distributional regression with covariate-dependent location, scale, and shape parameters.
- `poLCA`[9]. Polytomous latent class analysis for the categorical-indicator case; useful if baseline class indicators (e.g., symptom-checklist items) are incorporated into class assignment.
- `OpenMx` and `lavaan` with mixture extensions. Structural- equation mixture modelling for FMM-style specifications; more general but with steeper configuration cost than `lcmm` or `flexmix`.

Mplus is the reference implementation for GMM, FMM, and mixture-SEM; direct comparison with the identifiability literature occasionally requires access to it.

- Mplus is the standard environment for the psychometric mixture- model simulation literature on identifiability and overextraction.
- The Bauer-Curran, Lubke-Muthen, and Nylund studies all use Mplus.
- Reproducing published benchmarks may require Mplus access alongside the R implementations.

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The reference implementation outside R is Mplus [@mplus], which remains the standard environment for GMM, FMM, and related mixture-SEM specifications. Much of the published psychometric simulation literature on mixture-model identifiability and overextraction hazards [@bauercurran2003; @lubkemuthen2005; @nylund2007] uses Mplus, and direct comparison with that literature occasionally requires access to it.

13 Discussion

The biomarker-moderation problem sits at the intersection of biostatistical and psychometric traditions that have not previously engaged; recognising the encoding choice opens both sensitivity-analysis and analysis-strategy programmes.

- Biostatistics has efficient within-participant estimators but encodes moderation exclusively as a continuous interaction.
- Psychometrics has mature finite-mixture machinery applied only to cross-sectional settings well above the N-of-1 range.
- The continuous-versus-discrete choice is a substantive scientific question, not a parameterisation convenience.

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We begin by noting that the biomarker-treatment moderation problem in aggregated N-of-1 trials sits at the intersection of two methodological traditions that have, until now, only weakly engaged with each other. The biostatistical tradition has developed efficient estimators for within-participant treatment contrasts and the meta-analytic machinery to combine them, but has typically encoded biomarker moderation as a continuous interaction in fixed-effects regression. The psychometric tradition has developed mature finite-mixture and latent-class machinery for representing discrete subpopulation structure but has applied it primarily to cross-sectional or observational settings with sample sizes well above the N-of-1 range. The argument of this paper is that the choice between continuous-heterogeneity and latent-class encodings is a substantive scientific question, not a parameterisation convenience, and that recognising this choice opens both a sensitivity-analysis programme (the spectrum of biomarker-moderation specifications surveyed above) and an analysis-strategy programme (class-aware mixture analyses where identifiability per-

mits, and analysis-strategy mitigations where it does not).

Two limitations bound what the paper currently provides: the five-study simulation programme has not been executed, and the prazosin-PTSD re-analysis has not been carried out.

- The companion simulation plan describes but does not execute the empirical core of the present line of work.
- The prazosin-PTSD dataset is the most immediate real-data application target.
- Re-analysis should span at least three biomarker-moderation encodings and a class-aware analysis where convergence permits.

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Unfortunately, two limitations bound what the paper currently provides. First, the simulation programme that would empirically resolve the mixture-versus-mitigation efficiency question at trial-relevant sample sizes is described in the companion simulation plan but not yet executed. The five-study factorial outlined there (mixture-vs-MVN, the mean-covariance-combined triptych, heterogeneous random slopes, the three-regime breakdown grid, and the small-N identifiability and type I error pre-flight) constitutes the empirical core of the present line of work and is the next deliverable in the programme. Second, the application to real biomarker-guided trial data has not been carried out. The prazosin-PTSD analysis of @hendrickson2020 provides the most immediate target; the simulation programme should run in parallel with re-analysis of that dataset under at least three biomarker-moderation encodings and a class-aware analysis where convergence permits.

Three open questions remain: primary usability of class-aware analyses at small N, analysis half-life specification under a mixture DGP, and whether trial design should respond to the encoding question.

- Whether class-aware analyses are usable as primary tests or only as sensitivity checks requires direct simulation in the trial-relevant regime.
- The continuous-decay D_{it} has a clean single-component interpretation; its meaning under class-specific carryover dynamics is less obvious.
- If a latent-class DGP is biologically more plausible, designs minimising carryover may gain power the standard hybrid design surrenders.

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Three open methodological questions remain. The first is whether class-aware analyses are usable as primary analyses at any realistic N-of-1 sample size, or only as sensitivity checks. The identifiability evidence from the broader mixture-model literature suggests the latter; only direct simulation in the trial-relevant parameter regime can settle this. The second is how analysis half-life should be specified when the DGP is a mixture rather than a single MVN. The continuous-decay drug indicator $D_{it} = \exp(-\lambda t_{sd,it})$ *has a clean interpretation under a single-component generative model; its interpretation when the underlying generative process is class-structured and class-conditional carryover dynamics differ between classes is less obvious and may require analysis-side mixture extension. The third is whether the trial design itself should respond to the moderation-encoding question: if a latent-class generative form is biologically more plausible than a continuous one, designs that maximise the per-participant contrast and minimise carryover may gain power that the standard hybrid design surrenders.*

The five-study simulation programme addresses the open questions sequentially, with pre-registered ADEMP designs, uniform analysis wrappers, and scoped infrastructure additions.

- Factorial designs are pre-registered through ADEMP tables committed before production runs.
- Class-aware analyses are wrapped uniformly for direct cross-study comparability.
- The required DGP infrastructure additions are enumerated, scoped, and sequenced to support both this paper and the companion programme.

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The five-study simulation programme described in the companion plan is structured to address each of these questions sequentially. The factorial designs are pre-registered through ADEMP [@morris2019] tables committed before production runs; the class-aware analyses are wrapped uniformly so that simulation results are directly comparable across studies; and the infrastructure required to add the necessary DGPs to the existing simulation framework is enumerated, scoped, and sequenced to support both this paper's revisions and the companion carryover and biomarker-interaction papers in the wider research programme.

13.1 Why the analysis model is simpler than the data-generating process

The asymmetry between a structured DGP and a simple four-term analysis model is intentional: a component-matched analysis is impractical at

trial-relevant N.

- The DGP decomposes into BR, PB, and TV components; the analysis model is `Sx ~ bm + t + Dbc + bm:Dbc`.
- A component-by-component mirror would inflate the variance of $\beta_{bm:D}$ at $N \in [30, 150]$.
- Four reasons are given below for why the simpler analysis is preferred.

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A natural reaction to the simulation programme described above is that the DGP is structured (a three-component decomposition into biological response, placebo-belief response, and time-variant natural history under the standard pmsimstats parameterisation; or a latent-class mixture under the alternatives in this paper) while the analysis model is a four-term linear mixed model `Sx ~ bm + t + Dbc + bm:Dbc` with random intercept and AR(1) residual correlation. A reader newly encountering this asymmetry is reasonable to ask why the analysis does not mirror the DGP component-by-component, and whether doing so would improve inference about the moderation estimand. The short answer, with brief justification, is the following.

Component-matched analysis is impractical at trial-relevant N for four reasons: identification requirements, parameter inflation, convergence fragility, and robustness of the moderation estimand.

- Many designs supply only a subset of the contrasts needed to identify BR, PB, and TV components separately.
- Each Gompertz component adds 3–4 population parameters, inflating $\mathrm{Var}(\hat{\beta}_{bm:D})$ at $N \in [30, 150]$.
- $\beta_{bm:D}$ is robust to BR-versus-PB attribution, making the simpler model adequate for the validation question.

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A component-matched analysis (a non-linear mixed-effects model fitting Gompertz BR, PB, and TV components with participant-specific random effects) is impractical for trial-relevant N for four reasons: the trial design must supply the contrasts that identify each component (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV), and many designs supply only some of them; each Gompertz component costs *roughly three to four parameters at the population level and additional random-effect parameters per participant, which at $N \in [30, 150]$ inflates the variance of the moderation estimand more than it reduces residual confounding; the non-linear fitting machinery has higher convergence-failure rates and is more sensitive to starting values than the linear-mixed analogue; and the moderation estimand $\beta_{bm:D}$ is robust to the BR-versus-PB attribution for the validation question, even though it is not robust to the larger questions of whether the moderation is class-structured (the present paper) or covariance-versus-mean-encoded (the `dgp-mean-moderation-vs-mvn` companion).*

Adding a phase indicator is the minimal tractable extension that tests whether the apparent moderation shrinks under blinding.

- The phase-indicator extension tests whether the drug effect and its moderation shrink under blinding – a PB-mediation signature.
- For papers where the primary estimand is the marginal moderation, the simple four-term model is adequate.
- The phase-indicator extension is a candidate sensitivity analysis for Study 1 once the primary cells are run.

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The smallest extension that retains tractability while improving the BR-versus-PB sensitivity of the inference is a phase indicator plus its interaction with the drug indicator, schematically `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase + (1 | ptID)`. This adds three to four parameters and tests two things directly: whether the apparent drug effect shrinks under blinding (a signature that the open-label drug effect was partly PB-mediated) and whether the moderation itself shrinks under blinding (a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR). For papers where pharmacological-versus-placebo disentanglement is a secondary scientific question, this extension is the right step; for papers where the primary estimand is the marginal moderation, the simple `Sx ~ bm + t + Dbc + bm:Dbc` analysis is adequate. The simulation programme fixed in the companion plan uses the simple form throughout; the phase-indicator extension is a candidate sensitivity analysis to add to Study 1 (the mixture-vs-MVN factorial) once the primary cells are run. The document at `docs/24-component-decomposition-pedagogy.md` develops this point in more detail with worked numerical examples.

13.2 The latent-class DGP as a generator of apparent subadditivity

The latent-class DGP is a natural generator of apparent subadditivity in a one-component analysis; the $\gamma > 0$ finding is ambiguous between direct mechanistic interaction and emergent class structure.

- No direct $BR \times PB$ interaction is required in the class-conditional model to generate a non-zero $\hat{\gamma}$.
- Two mechanisms produce $\gamma > 0$: direct mechanistic coupling and emergent covariance from class structure.
- The companion BR-PB-TV decomposition paper defines and tests both mechanisms.

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A connection worth flagging explicitly: the latent-class DGP specified in Study 1 is a natural mechanism for generating apparent subadditivity in a downstream one-component analysis, even when no direct $BR \times PB$ interaction is specified in the class-conditional generative model. The companion methods paper at `analysis/report/06-component-decomposition/` defines subadditivity as a non-zero coefficient γ on a $BR \times PB$ interaction term in the conditional-mean specification of the additive decomposition, and notes that $\gamma > 0$ in a fitted additive-extension analysis is ambiguous between two distinct underlying mechanisms.

Emergent subadditivity arises when class membership correlates with both BR and PB strength, producing a non-zero marginal $E[BR \cdot PB]$ covariance term that mimics a mechanistic interaction.

- Responder-class participants having higher BR and PB on average induces a positive $E[BR \cdot PB]$ covariance via the unobserved class label.
- The magnitude depends on the gating function $\pi(B)$.
- From the additive-extension perspective this is indistinguishable from a γ-weighted mechanistic interaction.

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The first mechanism is direct mechanistic interaction: shared neural substrate, ceiling effects on the symptom scale, or belief-dynamics adaptation each produce a true non-additive coupling between drug and placebo response. The second mechanism is emergent subadditivity from latent-class structure: if the trial population is a mixture of responder and non-responder classes (the formulation in Section 3 of the present paper) and class membership correlates with both BR and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose *magnitude depends on the gating function $\pi(B)$. From the additive-extension analysis this looks exactly like a γ-weighted interaction.*

The Study 1 `pb_class_correlation` axis lets the analyst distinguish emergent from mechanistic subadditivity empirically within the simulation output.

- The axis crosses PB-class-independent and PB-class-correlated cells at matched marginal effect size.
- Large $\hat{\gamma}$ in the PB-class-correlated cell but small in the PB-class-independent cell identifies the emergent mechanism.
- The cross-paper synthesis with the BR-PB-TV decomposition paper directly falsifies or confirms class-aware versus mechanistic interpretations of apparent subadditivity.

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The Study 1 pre-registration accordingly includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels: a PB-class-independent cell (matching the original specification) and a PB-class-correlated cell (generating apparent subadditivity). The Study 1 simulation output therefore lets the analyst distinguish the two mechanisms empirically: if the additive-extension $\hat{\gamma}$ is large in the PB-class-correlated cell but small in the PB-class-independent cell at matched marginal effect size, the mechanism is emergent rather than mechanistic. The cross-paper synthesis with the BR-PB-TV decomposition paper is therefore not merely methodological convenience but a direct route to falsifying or confirming a class-aware versus mechanistic-interaction interpretation of any apparent subadditivity finding.

13.3 Progress to date

The five-study production programme is pre-registered under the ADEMP framework of Morris, White, and Crowther[10] at `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`; the present manuscript reports a Study 5 pilot at 240 replicates per cell.

Phase 1 of the simulation programme has reached the infrastructure-and-pilot milestone. Three components are committed and exercised end-to-end on the host R session.

Pre-registration. ADEMP tables for all five studies[10] are committed at `analysis/scripts/latent-class-mixture-application/ 00-ademp-pre-registration.md`, with cell counts (approximately 2,300 cells across the programme), seeded random-number control, an effect-size calibration sub-study, and a deviation-log specification. Subsequent production runs are diagnosed against these pre-registered estimands rather than against any post-hoc wrapper choice.

Phase 1 has committed three infrastructure components: ADEMP pre-registration, the `lcmm_analysis()` wrapper, and the heterogeneous-random-slopes DGP.

- ADEMP tables cover approximately 2,300 cells across the five studies with seeded RNG control and a deviation-log specification.
- The class-aware analysis wrapper uses a two-stage fit anchoring gridsearch starts from a single-class `hlme`.
- The heterogeneous-random-slopes DGP adds a Bernoulli class- membership channel with logistic gating in the standardised biomarker.

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Phase 1 of the simulation programme has reached the infrastructure-and-pilot milestone. Three components are committed and exercised end-to-end on the host R session. ADEMP tables for all five studies are committed with approximately 2,300 cells across the programme, seeded random-number control, an effect-size calibration sub-study, and a deviation-log specification. Subsequent production runs are diagnosed against these pre-registered estimands rather than against any post-hoc wrapper choice.

The `lcmm_analysis()` wrapper extracts two moderation signals: the primary gating-coefficient z-test and the secondary within-class $\beta_{bm:D}$ t-test.

- The two-stage fit anchors gridsearch starting values from a single-class `hlme` before fitting `hlme(ng=2)`.
- The gating-coefficient z-test identifies whether the biomarker predicts class membership.
- The within-class $\beta_{bm:D}$ t-test is retained for inferential symmetry with the linear-mixed baseline.

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Class-aware analysis wrapper. An `lcmm_analysis()` wrapper around `lcmm::hlme` is committed at `01-lcmm-wrapper.R` with the same input-output contract as the existing `lme_analysis()`. The wrapper uses a two-stage fit (a single-class `hlme` is fitted first to anchor starting values for a `gridsearch`-based multi-start `hlme(ng=2)` fit) and extracts two moderation signals: the gating-coefficient z-test on the biomarker (the *primary class-aware moderation test, identifying whether the biomarker predicts class membership) and the within-class $\beta_{bm:D}$ t-test (the secondary signal, retained for inferential symmetry with the linear-mixed baseline).*

Smoke-test results confirm the heterogeneous-random-slopes DGP produces the expected class structure and that `lcmm` discriminates it from the single-component MVN-DGP.

- Single-replicate test at $N = 70$, gating slope 1.5 produces 46% responders and marginal $\mathrm{Cor}(B, BR) = 0.63$.
- The linear-mixed `bm:Dbc` test detects the signal ($p = 0.002$) without class awareness.
- `lcmm` returns entropy 0.86 and BIC 170 lower than the matched single-component fit.

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Heterogeneous-random-slopes DGP. A wrapper around `generateData()` that adds a Bernoulli class-membership channel with logistic gating in the standardised biomarker and class-specific multipliers on the BR factor at on-drug timepoints is committed at `02-heterogeneous-slopes-dgp.R`. On a single-replicate smoke test ($N = 70$, gating slope 1.5, responder multiplier 1.0, non-responder multiplier 0.0) the DGP produces 46% responders and a marginal $\mathrm{Cor}(B, BR) = 0.63$ that the linear-mixed `bm:Dbc` test detects ($p = 0.002$) without class awareness; the `lcmm` fit returns a posterior class-assignment entropy of 0.86 and a BIC 170 lower than the matched single-component MVN-DGP fit, indicating that the mixture model discriminates the two DGPs at this parameter cell.

The 100-replicate Study 5 pre-flight pilot establishes a 32:1 per-replicate cost ratio for the class-aware versus linear-mixed analysis, with 96% `lcmm` convergence.

- Per-replicate timing: 0.05 s linear-mixed baseline; 1.57 s class-aware analysis.
- `lcmm` convergence rate: 96%; four failures arose from gridsearch initialisation.
- Results are checkpointed at `output/03-study5-pilot.rds`.

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Study 5 pre-flight pilot. A 100-replicate pilot under the single-component null DGP (Architecture B with $c_{bm} = 0$, $N = 70$, hybrid OL design) is reported at `output/03-study5-pilot.rds`. Per-replicate timing is 0.05 s for the linear-mixed baseline and 1.57 s for the class-aware analysis (32:1 cost ratio, consistent with the smoke-test estimate). `lcmm` convergence rate is 96%; the four non-converging replicates encountered numerical issues during the gridsearch initialisation.

Type I error at $\alpha = 0.05$ (Monte Carlo standard error 0.022) varies sharply across candidate test forms, motivating the choice of the primary class-aware moderation test:

Test	Rejection rate
lme `bm:Dbc` Wald t-test	0.000
lcmm gating-on-`bm` Wald z (unconditional)	**0.250**
lcmm within-class `bm:Dbc` Wald z	0.000
BIC selects ng=2 over ng=1	0.010
LRT chi-sq(df=4), naive reference	0.146
BIC-conditional gating procedure	0.000

The unconditional Wald-on-gating inflation is a finite-sample identifiability artefact: the conditional Wald SE underestimates true variability because the optimiser latches onto random class boundaries.

- Empirical SD of the gating coefficient (63.9) is fifty times the median within-replicate Wald SE (0.86).
- The gating coefficient is centred at zero with no systematic bias; inflation is in the SE, not the estimate.
- This is the overextraction artefact predicted by @bauercurran2003 and @lubkemuthen2005.

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The unconditional Wald-on-gating test rejects at five times the nominal rate. Diagnosis of the rejecting replicates shows that the empirical standard deviation of the gating coefficient across replicates (sd = 63.9) is roughly fifty times the median within-replicate Wald SE (0.86), and the empirical sd of the z-statistic is 1.47 against a nominal 1.0. The gating coefficient is centred at zero (median −0.10, sign balance 46/54) and shows no systematic bias; the inflation is driven by the conditional Wald SE under-estimating the true variability of $\hat{\beta}_{gating}$ at trial-relevant N. This is a finite-sample identifiability artefact predicted by the @bauercurran2003 and @lubkemuthen2005 overextraction literature: when the data have no class structure but `hlme` is forced to fit `ng=2`, the optimiser latches onto random patterns to define *a class boundary, and the resulting Wald SE does not absorb the labelling uncertainty.*

The naive LRT is anti-conservative because the true distribution at the $ng = 1$ boundary is chi-bar-squared, not chi-squared; the bootstrapped LRT is correct but too expensive.

- Rejection rate 0.146 under the naive chi-squared($df = 4$) reference.
- The true LRT distribution at the $ng = 1$ boundary is a chi-bar-squared mixture (McLachlan and Peel, ch. 6).
- The @nylund2007 bootstrapped LRT is the standard calibration but is too expensive at the class-aware simulation per-fit budget.

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The naive LRT against a chi-squared($df = 4$) reference is similarly anti-conservative (rejection 0.146), because the true LRT distribution at the `ng=1` boundary is a chi-bar-squared mixture rather than a chi-squared (@mclachlanpeel2000, ch. 6). The bootstrapped LRT of @nylund2007 is the standard calibration but is computationally expensive at the per-fit budget of a class-aware simulation.

The BIC-conditional gating procedure controls Type I at $N = 70$ under the null: BIC selects $ng = 2$ in 1% of replicates and none of those also reject the gating Wald.

- The procedure requires $\Delta_{\text{BIC}} > 0$ (or > 6 per Raftery 1995) before testing the gating coefficient.
- Zero conditional procedure rejections at the null gives Type I $= 0$ at the pilot's resolution.
- Whether the procedure has adequate power under a true two-class DGP is the central question for the rest of Study 5.

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The pragmatic class-aware moderation test is therefore a BIC-conditional gating procedure: fit `ng=1` and `ng=2`, prefer `ng=2` only if BIC supports it ($\Delta_{BIC} > 0$, or more conservatively > 6 per Raftery 1995), and only then test the gating coefficient. At $N = 70$ under the null DGP, BIC selects `ng=2` in 1% of replicates, and zero of those replicates also reject the gating Wald, giving the conditional procedure a Type I error of 0 at the pilot's resolution.

The conditional procedure is therefore usable as a primary moderation test at trial-relevant N. Whether the same conditional procedure has adequate power to detect a true two-class DGP is the central question for the rest of Study 5; pilots at non-null cells will measure this directly.

The earlier claim that class-aware analyses are not usable as primary tests at small N requires correction: it applies to the unconditional Wald test, not to the BIC-conditional procedure.

- The corrected proposition is that a single mixture-model coefficient's Wald test is unreliable at small N.
- A model-comparison wrapper (BIC or BLRT) is required to control Type I error.
- The BIC-conditional procedure is the pragmatic primary test at trial-relevant N.

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The diagnostic distinction matters for the paper's argument. The earlier proposition that 'class-aware analyses are not usable as primary at trial-relevant N' is too strong as stated: it applies to the unconditional Wald-on-gating test but not to the BIC-conditional procedure. The corrected proposition is that the unconditional Wald test on a single mixture-model coefficient is unreliable at small N and requires a model-comparison wrapper (BIC or BLRT) to control Type I error.

The full five-study programme is feasible on a single workstation in approximately one week of background compute at 8–16 cores.

- Study 1's class-aware arm requires approximately 340 serial hours, or 22–44 hours on 8–16 cores.
- Study 5 production is the largest block at approximately 60 hours per arm on 8 cores.
- The calibration sub-study and Study 5 pre-flight should run first to finalise effect-size axes before committing to Studies 1, 2, and 3.

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Production-budget implication. Extrapolated to the pre-registered cell counts (Study 1 has 768 cells × 1,000 replicates = 768,000 fits per analysis arm), the class-aware arm of Study 1 is approximately 340 hours of serial CPU time *at the pilot's mean fit time. Parallelisation across 8–16 cores reduces this to 22–44 hours per arm. Studies 2 and 3 require similar budgets; the Study 5 production run (5,000 replicates × three sample sizes × three class counts × three DGP families) is the largest single block at approximately 60 hours per arm on 8 cores. The full programme is feasible on a single workstation over a week of background compute, with the calibration sub-study and Study 5 pre-flight running first to finalise effect-size axes and identifiability cuts before committing to Studies 1, 2, and 3.*

The open question raised by the pilot is whether the BIC selection step has sufficient sensitivity under true class structure: Pilot 4 will test this across gating slopes and class-separation values.

- A conservative null-controlling procedure that rarely fires under true class structure provides no value.
- Expected pattern: BIC-conditional outperforms lme at large class separations; underperforms at moderate separations.
- Pilot 4 will fit the BIC-conditional procedure to the heterogeneous-slopes DGP from Study 3.

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Open question raised by the pilot. With Type I error of the BIC-conditional procedure controlled at the null at $N = 70$, the next-most-pressing identifiability question is whether the BIC selection step has enough sensitivity to detect a real two-class DGP at trial-relevant N. A class-aware procedure that is conservative under the null but rarely fires under true class structure is no more useful than a procedure that fires too often. Pilot 4 (planned for the next iteration) will fit the same BIC-conditional procedure to the heterogeneous-random-slopes DGP from Study 3 (with several values of the gating slope and class-separation Δ) and report the power profile of the conditional procedure alongside that of the linear-mixed baseline. We expect that the BIC-conditional procedure will outperform the linear-mixed `bm:Dbc` test at large class separations (where `ng=2` BIC consistently wins) but will underperform it at moderate class separations (where BIC is undecided and the procedure falls back to a default that does not exploit the latent-class structure).

13.4 Pilot results (Study 5, 240 replicates)

The 240-replicate Study 5 pilot crossed three gating-slope levels and three test forms at $N = 70$, with MCSE approximately 0.030 at working power.

- Wall time 558 s on 4-core `parallel::mclapply`.
- `lcmm` convergence: 92.9% at the null, 98.3–98.8% at non-null cells.
- MCSE approximately 0.030 at working power near 0.5.

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The Study 5 prototype was rerun at 240 replicates per cell on 4-core `parallel::mclapply` with three gating-slope levels (gs $\in \{0, 1.0, 1.5\}$) and three test forms (lme `bm:Dbc` Wald, `lcmm` gating Wald unconditional, BIC-conditional gating). Wall time was 558 s; convergence was 92.9% under the null and 98.3–98.8% at non-null cells. Per-replicate output is at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP summary at `03-latent-summary.txt`. Monte Carlo standard error at the working power near 0.5 is approximately 0.030.

The unconditional Wald inflates Type I to 0.197; the BIC- conditional procedure controls Type I but is severely under-powered, approximating a never-reject rule under heterogeneous slopes.

- Unconditional Wald Type I 0.197 (95% CI $[0.145, 0.250]$), decisively above the nominal 0.05; power below the lme baseline at both non-null cells.
- BIC-conditional Type I 0.004 under the null; power 0.025 at gs $= 1.0$ and 0.004 at gs $= 1.5$.
- Non-monotonicity at gs $= 1.5$ suggests the lme captures the heterogeneous-slopes signal well enough that BIC continues to favour $ng = 1$.

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The unconditional Wald-on-gating type I error of 0.197 (95% binomial CI $[0.145, 0.250]$) is now decisively distinguishable from the nominal 0.05 (more than five MCSE of separation) and is statistically consistent with the 100-replicate pilot's 0.250 point estimate. Power at $gs = 1.0$ (0.304) and $gs = 1.5$ (0.360) sits below the lme baseline (0.325 and 0.542 respectively), so the unconditional Wald buys nothing in exchange for the type I inflation. The BIC-conditional procedure does control type I (0.004 under the null, with upper CI near 0.013), but its power is 0.025 at $gs = 1.0$ and 0.004 at $gs = 1.5$: under the heterogeneous-random-slopes DGP at $N = 70$, the BIC step selects `ng=1` in essentially every replicate and the conditional procedure approximates a never-reject rule. The non-monotonicity ($gs = 1.5$ below $gs = 1.0$) suggests that as the heterogeneous-slopes signal strengthens, a single-class lme captures it well enough that BIC continues to favour `ng=1` and the conditional procedure does not fire.

The lme `bm:Dbc` test dominates both `lcmm` tests at this N and DGP

regime; a canonical 1,000-replicate run is required to map the regime where the BIC-conditional procedure becomes competitive.

- lme holds nominal Type I (0 of 240) and reaches 0.542 power at gs $= 1.5$.
- Neither class-aware test is competitive: the unconditional test is anti-conservative; the BIC-conditional test is severely under-powered.
- A canonical run at $N \in \{35, 70, 100\}$ with a wider gating-slope grid is required to locate the competitive regime, plausibly at large class separations.

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The lme `bm:Dbc` test holds nominal type I (0 of 240 under the null) and reaches 0.542 power at $gs = 1.5$, dominating both `lcmm` tests at this sample size and DGP regime. We therefore conclude, on the evidence of this pilot, that at trial-relevant N and under the heterogeneous-random-slopes generating mechanism examined here, neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline: the unconditional test is severely anti-conservative, and the BIC-conditional test is severely under-powered. A canonical run at 1,000 replicates per cell with a wider gating-slope grid (and at $N \in \{35, 70, 100\}$) is required to map the regime in which the BIC-conditional procedure does become competitive, plausibly at large class separations rather than under heterogeneous random slopes.

14 Conflict of interest

The authors declare no conflicts of interest.

15 Funding

This work received no specific funding.

16 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

17 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (4-core `mclapply` for the `lcmm`-bottlenecked run), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The class-aware analyses additionally use `lcmm` with `gridsearch` initialisation.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs from the 240-replicate Study 5 pilot are checkpointed at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP cell-level summaries at `analysis/data/quick-sim/03-latent-summary.txt`; the effect-size calibration table at `analysis/data/quick-sim/03-calibration-table.rds`. Driver scripts are under `analysis/scripts/latent-class-mixture-application/`, including the pilot driver `03-study5-pilot.R` (executed) and the production driver `10-study5-production.R` (committed; the five-study production programme has not yet been run). The pre-registered ADEMP plan is at `analysis/scripts/latent-class-mixture-application/ 00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/03-latent-class-mixture-application/`.

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